

Module - 04

Enumeration

→ Enumeration: It is the process of actively extracting detailed information from a target. Deeper reconnaissance by interacting with the target.

→ Reconnaissance vs. Enumeration:

Reconnaissance	Enumeration
Discovers the target	Queries the target for details
Often passive	Usually active
WhoIs, Google, OSINT	Nmap service detection, SMB, SNMP, LDAP queries.
Finds assets	Finds detailed information about assets.

→ Common Enumeration Examples:

- **DNS Enumeration**

- Enumerate DNS records (A, AAAA, MX, NS, TXT, CNAME)

- **SMTP Enumeration**

- **Service Enumeration**

- Determine service versions

- Identify supported protocols and configurations

- **SMB Enumeration**

- List shared folders

- Gather server information (on systems you are authorized to assess)

NetBIOS Enumeration

- BIOS: Basic Input/Output System, it's a firmware that checks hardware (CPU, RAM, SSD) working and load Bootloader.
- NetBIOS: Network Basic Input/Output System, network protocol that allows computer on LAN to communicate & share resources.

Service	Protocol	Port	Description
Name Service	UDP	137	Registers and Resolves computer names. [DNS] - name resolution.
Datagram Service	UDP	138	Used for Broadcasts.
Session Service	TCP	139	Used for file and printer sharing over NetBIOS. [SMB] - over TCP for file sharing.

- NetBIOS Enumeration: is the process of collecting information exposed by NetBIOS-enabled Windows systems.
- Samba: is an open-source software that allows Linux and Unix systems to communicate with Windows systems using SMB (Server Message Block) protocol.

NetBIOS Enumeration - nmap

- Target Machine: Metasploitable2

```
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:3d:78:90
          inet addr:192.168.110.130  Bcast:192.168.110.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe3d:7890/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:12499 errors:1 dropped:2 overruns:0 frame:0
          TX packets:9340 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:824917 (805.5 KB)  TX bytes:847173 (827.3 KB)
          Interrupt:17 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:551 errors:0 dropped:0 overruns:0 frame:0
          TX packets:551 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:244305 (238.5 KB)  TX bytes:244305 (238.5 KB)
```

- Command: `nmap -p 137,138,139 192.168.110.130`
Scanning NetBIOS Services.

```
(root@kali)-[~]
# nmap -p 137,138,139 192.168.110.130
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-02 07:33 -0400
Nmap scan report for 192.168.110.130
Host is up (0.0010s latency).

PORT      STATE SERVICE
137/tcp   closed netbios-ns
138/tcp   closed netbios-dgm
139/tcp   open  netbios-ssn
MAC Address: 00:0C:29:3D:78:90 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.66 seconds
```

- Command: nmblookup -A 192.168.110.130

[137/UDP]

-A = Query by IP Address.

```
(root@kali)-[~]
# nmblookup -A 192.168.110.130
Looking up status of 192.168.110.130
    METASPLOITABLE <00> - B <ACTIVE>
    METASPLOITABLE <03> - B <ACTIVE>
    METASPLOITABLE <20> - B <ACTIVE>
    .. __MSBROWSE__ . <01> - <GROUP> B <ACTIVE>
    WORKGROUP <00> - <GROUP> B <ACTIVE>
    WORKGROUP <1d> - B <ACTIVE>
    WORKGROUP <1e> - <GROUP> B <ACTIVE>

    MAC Address = 00-00-00-00-00-00
```

Field	Output	Meaning
Computer Name	METASPLOITABLE <00>	NetBIOS computer name.
File Server	METASPLOITABLE <20>	Indicates the host is running an SMB file server (Samba).

- Nmblookup: It is a Samba utility used to query the NetBIOS Name Service.

▪ Command: `nmap -sC -p 139 192.168.110.130`

[139/TCP]

Scanning Session Service using Script.

```
(root@kali)-[~]
# nmap -sC -p 139 192.168.110.130
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-02 07:36 -0400
Nmap scan report for 192.168.110.130
Host is up (0.0016s latency).

PORT      STATE SERVICE
139/tcp   open  netbios-ssn
MAC Address: 00:0C:29:3D:78:90 (VMware)

Host script results:
| smb-os-discovery:
|   OS: Unix (Samba 3.0.20-Debian)
|   Computer name: metasploitable
|   NetBIOS computer name:
|   Domain name: localdomain
|   FQDN: metasploitable.localdomain
|_  System time: 2026-06-30T12:36:59-04:00
|_ clock-skew: mean: -1d16h59m48s, deviation: 2h49m42s, median: -1d18h59m48s
| smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: disabled (dangerous, but default)
|_ smb2-time: Protocol negotiation failed (SMB2)
|_ nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)

Nmap done: 1 IP address (1 host up) scanned in 27.14 seconds
```

Field	Output	Meaning
Port Status	139/tcp open netbios-ssn	NetBIOS Session Service (SMB communication over NetBIOS) is running.
OS	Samba (3.0.20-Debian)	Host is running Samba 3.0.20 on Debian. This is old Samba Version.
Domain	localdomain	Workgroup/domain name.
Authentication level	user	User-level authentication is configured.
SMB Version	SMB2 negotiation failed.	Only SMB1 is supported.
NetBIOS Name	METASPLOITABLE	NetBIOS hostname of the system.

NetBIOS Enumeration - enum4linux

→enum4linux: Linux enumeration tool that gather information from Windows and Samba (SMB) systems.

- All-in-one SMB/NetBIOS information gathering tool.

- Command: enum4linux 192.168.110.130

```
root@kali: ~  root@kali: ~ 
(root@kali)-[~]
# enum4linux 192.168.110.130
Starting enum4linux v0.9.1 ( http://labs.portcullis.co.uk/application/enum4linux/ ) on Fri Jul 3 03:07:27 2026

===== ( Target Information ) =====
Target ..... 192.168.110.130
RID Range ..... 500-550,1000-1050
Username ..... ''
Password ..... ''
Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none

===== ( Enumerating Workgroup/Domain on 192.168.110.130 ) =====

[+] Got domain/workgroup name: WORKGROUP

===== ( Nbtstat Information for 192.168.110.130 ) =====

Looking up status of 192.168.110.130
METASPLOITABLE <00> - B <ACTIVE> Workstation Service
METASPLOITABLE <03> - B <ACTIVE> Messenger Service
METASPLOITABLE <20> - B <ACTIVE> File Server Service
.._MSBROWSE_.. <01> - <GROUP> B <ACTIVE> Master Browser
WORKGROUP <00> - <GROUP> B <ACTIVE> Domain/Workgroup Name
WORKGROUP <1d> - B <ACTIVE> Master Browser
WORKGROUP <1e> - <GROUP> B <ACTIVE> Browser Service Elections

MAC Address = 00-00-00-00-00-00

===== ( Session Check on 192.168.110.130 ) =====

[+] Server 192.168.110.130 allows sessions using username '', password ''

===== ( Getting domain SID for 192.168.110.130 ) =====

Domain Name: WORKGROUP
Domain Sid: (NULL SID)

[+] Can't determine if host is part of domain or part of a workgroup

===== ( OS information on 192.168.110.130 ) =====

[E] Can't get OS info with smbclient

[+] Got OS info for 192.168.110.130 from srvinfo:
METASPLOITABLE Wk Sv PrQ Unx NT SNT metasploitable server (Samba 3.0.20-Debian)
platform_id : 500
os version : 4.9
server type : 0x9a03
```




Session Actions Edit View Help

root@kali: ~

root@kali: ~

(Users on 192.168.110.130)

index	RID	Account	Name	Desc
index: 0x1	RID: 0x3f2	Account: games	Name: games	Desc: (null)
index: 0x2	RID: 0x1f5	Account: nobody	Name: nobody	Desc: (null)
index: 0x3	RID: 0x4ba	Account: bind	Name: (null)	Desc: (null)
index: 0x4	RID: 0x402	Account: proxy	Name: proxy	Desc: (null)
index: 0x5	RID: 0x4b4	Account: syslog	Name: (null)	Desc: (null)
index: 0x6	RID: 0xbba	Account: user	Name: just a user,111,,	Desc: (null)
index: 0x7	RID: 0x42a	Account: www-data	Name: www-data	Desc: (null)
index: 0x8	RID: 0x3e8	Account: root	Name: root	Desc: (null)
index: 0x9	RID: 0x3fa	Account: news	Name: news	Desc: (null)
index: 0xa	RID: 0x4c0	Account: postgres	Name: PostgreSQL administrator,,,	Desc: (null)
index: 0xb	RID: 0x3ec	Account: bin	Name: bin	Desc: (null)
index: 0xc	RID: 0x3f8	Account: mail	Name: mail	Desc: (null)
index: 0xd	RID: 0x4c6	Account: distccd	Name: (null)	Desc: (null)
index: 0xe	RID: 0x4ca	Account: proftpd	Name: (null)	Desc: (null)
index: 0xf	RID: 0x4b2	Account: dhcp	Name: (null)	Desc: (null)
index: 0x10	RID: 0x3ea	Account: daemon	Name: daemon	Desc: (null)
index: 0x11	RID: 0x4b8	Account: sshd	Name: (null)	Desc: (null)
index: 0x12	RID: 0x3f4	Account: man	Name: man	Desc: (null)
index: 0x13	RID: 0x3f6	Account: lp	Name: lp	Desc: (null)
index: 0x14	RID: 0x4c2	Account: mysql	Name: MySQL Server,,,	Desc: (null)
index: 0x15	RID: 0x43a	Account: gnats	Name: Gnats Bug-Reporting System (admin)	Desc: (null)
index: 0x16	RID: 0x4b0	Account: libuuid	Name: (null)	Desc: (null)
index: 0x17	RID: 0x42c	Account: backup	Name: backup	Desc: (null)
index: 0x18	RID: 0xbb8	Account: msfadmin	Name: msfadmin,,,	Desc: (null)
index: 0x19	RID: 0x4c8	Account: telnetd	Name: (null)	Desc: (null)
index: 0x1a	RID: 0x3ee	Account: sys	Name: sys	Desc: (null)
index: 0x1b	RID: 0x4b6	Account: klog	Name: (null)	Desc: (null)
index: 0x1c	RID: 0x4bc	Account: postfix	Name: (null)	Desc: (null)
index: 0x1d	RID: 0xbbc	Account: service	Name: ,,,	Desc: (null)
index: 0x1e	RID: 0x434	Account: list	Name: Mailing List Manager	Desc: (null)
index: 0x1f	RID: 0x436	Account: irc	Name: ircd	Desc: (null)
index: 0x20	RID: 0x4be	Account: ftp	Name: (null)	Desc: (null)
index: 0x21	RID: 0x4c4	Account: tomcat55	Name: (null)	Desc: (null)
index: 0x22	RID: 0x3f0	Account: sync	Name: sync	Desc: (null)
index: 0x23	RID: 0x3fc	Account: uucp	Name: uucp	Desc: (null)

user:[games] rid:[0x3f2]
user:[nobody] rid:[0x1f5]
user:[bind] rid:[0x4ba]
user:[proxy] rid:[0x402]
user:[syslog] rid:[0x4b4]
user:[user] rid:[0xbba]
user:[www-data] rid:[0x42a]
user:[root] rid:[0x3e8]
user:[news] rid:[0x3fa]
user:[postgres] rid:[0x4c0]
user:[bin] rid:[0x3ec]
user:[mail] rid:[0x3f8]
user:[distccd] rid:[0x4c6]
user:[proftpd] rid:[0x4ca]
user:[dhcp] rid:[0x4b2]
user:[daemon] rid:[0x3ea]
user:[sshd] rid:[0x4b8]
user:[man] rid:[0x3f4]
user:[lp] rid:[0x3f6]
user:[mysql] rid:[0x4c2]
user:[gnats] rid:[0x43a]
user:[libuuid] rid:[0x4b0]
user:[backup] rid:[0x42c]
user:[msfadmin] rid:[0xbb8]
user:[telnetd] rid:[0x4c8]
user:[sys] rid:[0x3ee]
user:[klog] rid:[0x4b6]
user:[postfix] rid:[0x4bc]
user:[service] rid:[0xbbc]
user:[list] rid:[0x434]
user:[irc] rid:[0x436]
user:[ftp] rid:[0x4be]
user:[tomcat55] rid:[0x4c4]

```
( Share Enumeration on 192.168.110.130 )

  Sharename      Type      Comment
  -----
  print$         Disk      Printer Drivers
  tmp            Disk      oh noes!
  opt            Disk
  IPC$           IPC       IPC Service (metasploitable server (Samba 3.0.20-Debian))
  ADMIN$         IPC       IPC Service (metasploitable server (Samba 3.0.20-Debian))
Reconnecting with SMB1 for workgroup listing.

  Server          Comment
  -----
  Workgroup        Master
  WORKGROUP        METASPLOITABLE

[+] Attempting to map shares on 192.168.110.130

//192.168.110.130/print$      Mapping: DENIED Listing: N/A Writing: N/A
//192.168.110.130/tmp        Mapping: OK Listing: OK Writing: N/A
//192.168.110.130/opt        Mapping: DENIED Listing: N/A Writing: N/A

[E] Can't understand response:
NT_STATUS_NETWORK_ACCESS_DENIED listing \*
//192.168.110.130/IPC$      Mapping: N/A Listing: N/A Writing: N/A
//192.168.110.130/ADMIN$    Mapping: DENIED Listing: N/A Writing: N/A

( Password Policy Information for 192.168.110.130 )

Password: █
```

Important Finding:

Field	Detail	Meaning
Session Check	Server allows sessions using username '', password ''	The SMB server allows a null session (anonymous login) .
Share Enumeration Share Access Map Shares	Tmp Mapping: OK Listening: OK	Can Connect to tmp share and list its contents.

- Command: `smbclient //192.168.110.130/tmp -N`
Smbclient = SMB client program.
/tmp = SMB share that we access.
-N = Null Session

```
(root@kali)-[~]
# smbclient //192.168.110.130/tmp -N
Anonymous login successful
Try "help" to get a list of possible commands.
smb: \> ls
.                D            0   Tue Jun 30 15:33:20 2026
..               DR            0   Sun May 20 14:36:12 2012
.ICE-unix        DH            0   Tue Jun 30 10:38:29 2026
.X11-unix        DH            0   Tue Jun 30 10:38:16 2026
5169.jsvc_up     R             0   Tue Jun 30 10:38:20 2026
.X0-lock        HR           11   Tue Jun 30 10:38:16 2026

7282168 blocks of size 1024. 5436820 blocks available
smb: \> put file.txt
putting file file.txt as \file.txt (0.0 kB/s) (average 0.0 kB/s)
smb: \> ls
.                D            0   Tue Jun 30 15:39:28 2026
..               DR            0   Sun May 20 14:36:12 2012
.ICE-unix        DH            0   Tue Jun 30 10:38:29 2026
.X11-unix        DH            0   Tue Jun 30 10:38:16 2026
5169.jsvc_up     R             0   Tue Jun 30 10:38:20 2026
.X0-lock        HR           11   Tue Jun 30 10:38:16 2026
file.txt        A             0   Tue Jun 30 15:39:28 2026

7282168 blocks of size 1024. 5436820 blocks available
smb: \> █
```

- The tmp SMB share is accessible anonymously.
- We can browse its contents.
- We can upload files to it.

LDAP | SNMP | NTP

- LDAP: Lightweight Directory Access Protocol, used to store, organize and retrieve information from directory service.
- SNMP: Simple Network Management Protocol, used to manage and monitor devices. Ex - Routers, Servers, IOT devices, Printer etc.
- NTP: Network Time Protocol, is a protocol used to synchronize the clocks of computers and network devices over a network.

Feature	SNMP	LDAP	NTP
Full Form	Simple Network Management Protocol	Lightweight Directory Access Protocol	Network Time Protocol
Purpose	Monitor and manage network devices	Manage and query directory information	Synchronize system time
Default Port	UDP 161	TCP 389 (LDAP), TCP 636 (LDAPS)	UDP 123
Works On	Routers, switches, servers, printers	Active Directory, OpenLDAP	Almost all network devices
Main Goal	Network monitoring and management	Centralized identity management	Accurate time synchronization

SMTP Enumeration

→SMTP: Simple Mail Transfer Protocol, used to send emails.

→SMTP Enumeration: the process of gathering information from SMTP server.

Port No:

Port no	Description
25	Default, Mail Transfer.
465	Legacy environment, uses SSL/TLS encryption.
587	Modern, uses STARTTLS encryption.

- Command: `nmap -p 25 192.168.110.130`

```
(root@kali)-[~]
# nmap -p 25 192.168.110.130
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-03 06:02 -0400
Nmap scan report for 192.168.110.130
Host is up (0.0019s latency).

PORT      STATE SERVICE
25/tcp    open  smtp
MAC Address: 00:0C:29:3D:78:90 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 4.72 seconds
```

- Command: `nmap -p 25 --script smtp-* 192.168.110.130`

```
(root@kali)-[~]
# nmap -p 25 --script smtp-* 192.168.110.130
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-03 06:46 -0400
Nmap scan report for 192.168.110.130
Host is up (0.00073s latency).

PORT      STATE SERVICE
25/tcp    open  smtp
|_smtp-open-relay: Server doesn't seem to be an open relay, all tests failed
|_smtp-vuln-cve2010-4344:
|_  The SMTP server is not Exim: NOT VULNERABLE
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
|_smtp-enum-users:
|_  Method RCPT returned a unhandled status code.
MAC Address: 00:0C:29:3D:78:90 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 32.89 seconds
```

Finding	Result	What it Means
SMTP Port	25/tcp open smtp	SMTP service is running.
Hostname	metasploitable.localdomain	Mail server hostname.
CVE	Cve2010-4344	Vulnerable

- Command: `nc 192.168.110.130 25`
nc = Netcat

```
(root@kali)-[~]
# nc 192.168.110.130 25
220 metasploitable.localdomain ESMTP Postfix (Ubuntu)
VRFY root
252 2.0.0 root
^Z
zsh: suspended nc 192.168.110.130 25
```

- Command: `telnet 192.168.110.130 25`

```
(root@kali)-[~]
# telnet 192.168.110.130 25
Trying 192.168.110.130 ...
Connected to 192.168.110.130.
Escape character is '^]'.
220 metasploitable.localdomain ESMTP Postfix (Ubuntu)
VRFY msfadmin
252 2.0.0 msfadmin
VRFY root
252 2.0.0 root
```

SMB Enumeration

→ **SMB:** SMB (Server Message Block) is a network protocol used to share files, printers, access remote directories and communication between computers on network.

- **Command:** `nmap -sC -p 139,445 192.168.110.130`

```
(root@kali)-[~]
# nmap -sC -p 139,445 192.168.110.130
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-03 07:28 -0400
Nmap scan report for 192.168.110.130
Host is up (0.00061s latency).

PORT      STATE SERVICE
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
MAC Address: 00:0C:29:3D:78:90 (VMware)

Host script results:
| smb-os-discovery:
|   OS: Unix (Samba 3.0.20-Debian)
|   Computer name: metasploitable
|   NetBIOS computer name:
|   Domain name: localdomain
|   FQDN: metasploitable.localdomain
|_  System time: 2026-06-30T18:58:32-04:00
|_ nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
|_ clock-skew: mean: -2d10h30m08s, deviation: 2h49m42s, median: -2d12h30m08s
|_ smb2-time: Protocol negotiation failed (SMB2)
| smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: disabled (dangerous, but default)

Nmap done: 1 IP address (1 host up) scanned in 32.89 seconds
```

Field	Output	Meaning
Port 139	open netbios-ssn	NetBIOS Session Service is running.
Port 445	open microsoft-ds	SMB service is running.
Authentication	User	User-level authentication is used.
Guest Access	account_used: guest	Nmap connected using the guest account.
OS	Samba (3.0.20-Debian)	Host is running Samba 3.0.20 on Debian. This is old Samba Version.

- Command: `smbclient -L //192.168.110.130 -N`
 -L = List all available SMB shares
 -N = Null / Anonymous login.

```
(root@kali)-[~]
# smbclient -L //192.168.110.130 -N
Anonymous login successful
```

Sharename	Type	Comment
print\$	Disk	Printer Drivers
tmp	Disk	oh noes!
opt	Disk	
IPC\$	IPC	IPC Service (metasploitable server (Samba 3.0.20-Debian))
ADMIN\$	IPC	IPC Service (metasploitable server (Samba 3.0.20-Debian))

```
Reconnecting with SMB1 for workgroup listing.
Anonymous login successful
```

Server	Comment
Workgroup	Master
WORKGROUP	METASPLOITABLE

- Command: `smbclient //192.168.110.130/tmp -N`

```
(root@kali)-[~]
# smbclient //192.168.110.130/tmp -N
Anonymous login successful
Try "help" to get a list of possible commands.
smb: \> help
```

?	allinfo	altname	archive	backup
blocksize	cancel	case_sensitive	cd	chmod
chown	close	del	deltree	dir
du	echo	exit	get	getfacl
geteas	hardlink	help	history	iosize
lcd	link	lock	lowercase	ls
l	mask	md	mget	mkdir
mkfifo	more	mput	newer	notify
open	posix	posix_encrypt	posix_open	posix_mkdir
posix_rmdir	posix_unlink	posix_whoami	print	prompt
put	pwd	q	queue	quit
readlink	rd	recurse	reget	rename
reput	rm	rmdir	showacl	setea
setmode	scopy	stat	symlink	tar
tarmode	timeout	translate	unlock	volume
vuid	wdel	logon	listconnect	showconnect
tcon	tdis	tid	utimes	logoff
..	!			

```
smb: \>
```